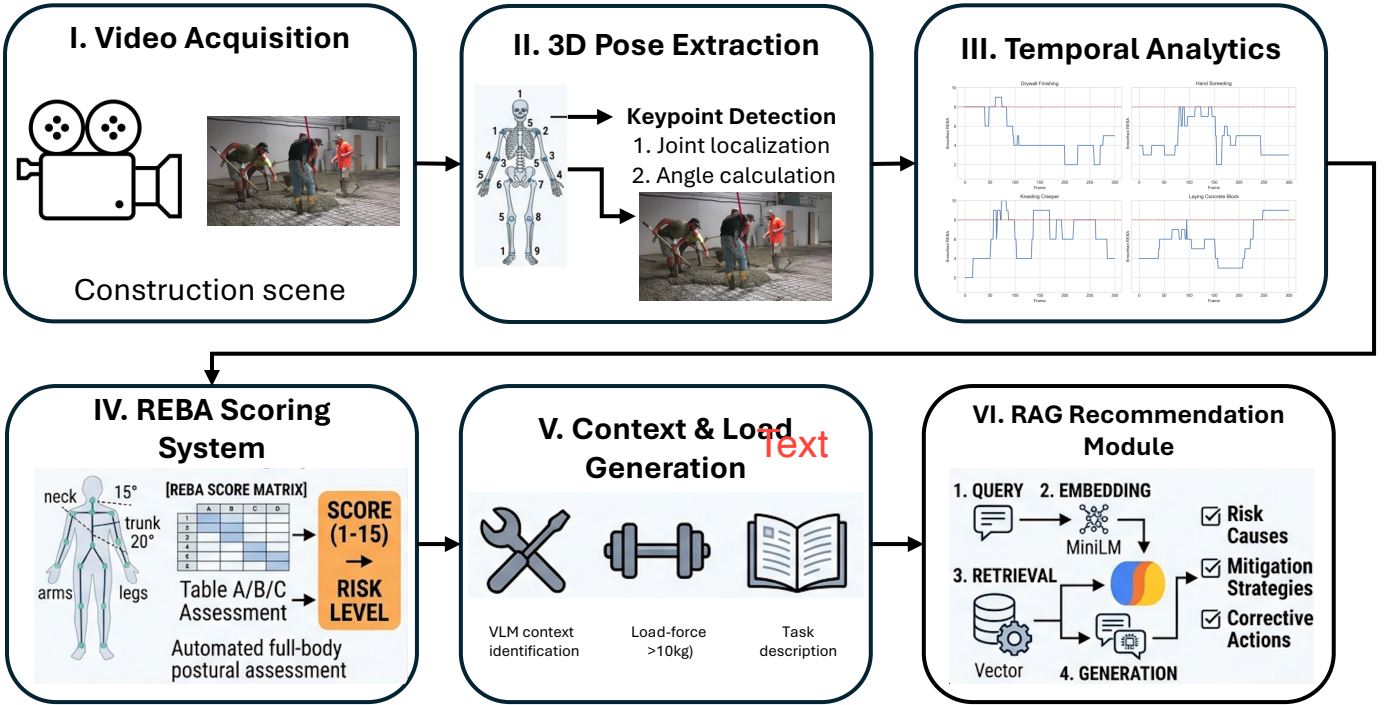
I. Video Acquisition
Construction scene
II. 3D Pose Extraction
Keypoint Detection
1. Joint localization
2. Angle calculation
III. Temporal Analytics
Drywall Finishing
Hand Screeding
Kneading Creeper
Laying Concrete Block
IV. REBA Scoring System
neck
15°
trunk
20°
arms
legs
[REBA SCORE MATRIX]
Table A/B/C Assessment
SCORE (1-15)
RISK LEVEL
Automated full-body postural assessment
V. Context & Load Generation
Text
VLM context identification
Load-force >10kg)
Task description
VI. RAG Recommendation Module
1. QUERY
2. EMBEDDING
MiniLM
3. RETRIEVAL
Vector
4. GENERATION
Risk Causes
Mitigation Strategies
Corrective Actions